Homework 3 – Regression

1 Question 1

Below, I simulate a rather quite annoying set of regression data.

```r
set.seed(0510)
n <- 250
dat.q1 <- tibble(x1 = rnorm(n, mean = 50, sd = 10),
                 x2 = factor(sample(c("A", "B"), n, replace = TRUE)),
                 x3 = factor(sample(c("Control", "Treat"), n, replace = TRUE)),
                 x4 = factor(sample(c("Low", "High"), n, replace = TRUE), levels=c("Low", "High")),
                 x5 = rnorm(n, mean = 0, sd = 1),
                 x6 = factor(sample(c("Group1", "Group2"), n, replace = TRUE)),
                 x7 = rnorm(n, mean = 10, sd = 2),
                 x8 = rnorm(n, mean = 10, sd = 2)) |>
        mutate(y = 5 +                          # Intercept
                 # Quantiative x1
                 2 * x1 +
                 # Categorical x2
                 (x2 == "B") * 10 +
                 # Categorical x Categorical (x3*x4)
                 (x3 == "Treat") * 5 + (x4 == "High") * 2 +
                 (x3 == "Treat" & x4 == "High") * 15 +
                 # Quantitative x Categorical  (x5*x6)
                 3 * x5 + (x6 == "Group2") * 4 +
                 (x5 * (x6 == "Group2")) * 8 +
                 # Quantitative x Quantitative (x7*x8)
                 1.5 * x7 + 0.5 * x8 +
                 0.8 * (x7 * x8) +
                 # Error
                 rnorm(n, mean = 0, sd = 15))
```

1.1 Part a

Fit the regression model.

$$
\begin{aligned}
Y_i = &\beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \\
&\beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\
&\beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + && \text{(Categorical} \times \text{Categorical)} \\
&\beta_{10} (x_{5i} \times I(x_{6i} = \text{Group2})) + && \text{(Continuous} \times \text{Categorical)} \\
&\beta_{11} (x_{7i} \times x_{8i}) + \epsilon_i && \text{(Continuous} \times \text{Continuous)}
\end{aligned}
$$

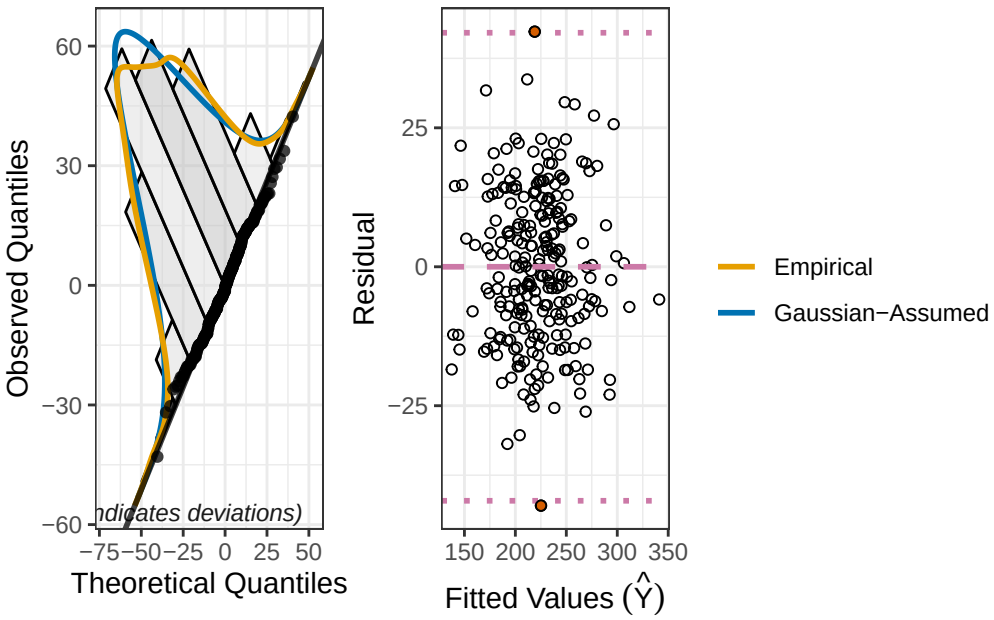
Solution: First, we fit the regression model using `lm()`.

```r
mod.q1 <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.q1)
```

1.2 Part b

Conduct a residual and outlier analysis.

```r
plotResiduals(mod.q1)
```



Using `plotResiduals()` from the included *plotResiduals.r* file, I can see that the residuals seem to follow a normal distribution and there are few outliers (only two). Additionally the residual plot on the right shows constant variance across the fitted values and the residuals are centered around 0.

1.3 Part c

Interpret the intercept, the coefficients on x_1 and x_2.

```r
knitr::kable(mod.q1$coefficients[1:3])
```

	x
(Intercept)	8.896532
x1	2.135460
x2B	10.028512

The intercept (8.9) is the predicted value of the dependent variable if all of the dependent variables are 0. x_1 is a continuous value so the coefficient for x_1 (2.14) is how much the predicted value of the dependent variable changes when x_1 changes and all other independent variables are held constant. So in this case if all dependent variables are held constant and x_1 is increased by 1 then $\hat{y}$ increases by (2.14). x_2, however, is a categorical value so the interpretation of its coefficient is different. There are only two possible values of x_2 'A' or 'B', and the variable in the regression model is labeled x_2B. This tells us that when x_2 is 'A' $x_2B = 0$ and when x_2 is 'B' $x_2B = 1$, so when x_2 is 'A' $\hat{y}$ does not change but when x_2 is 'B' $\hat{y}$ increases by 10.03.

1.4 Part d

Untangle and interpret the $x_3 * x_4$ interaction using algebra. Use Table 1 and Figure 1 for context.

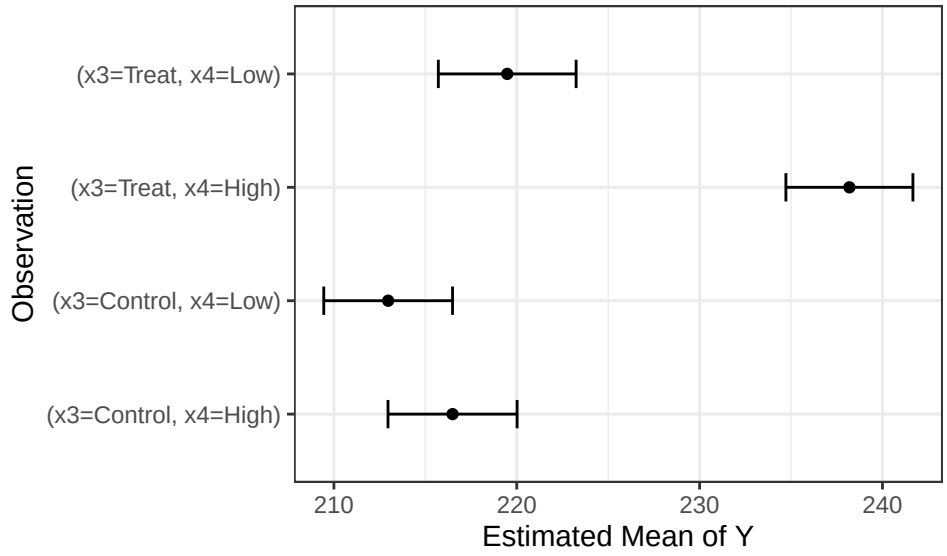
```r
library(emmeans)
emm.q1d <- emmeans(mod.q1, ~x3*x4) |> # Don't average across levels: use |x2*x6
            data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")
```

Table 1: Estimated marginal means for the model in Question 1.

x3	x4	Mean	SE	df	Lower	Upper
Control	Low	212.9676	1.7871	238	209.4470	216.4882
Treat	Low	219.4792	1.9123	238	215.7120	223.2464
Control	High	216.4894	1.7916	238	212.9601	220.0188
Treat	High	238.1930	1.7631	238	234.7198	241.6662

```
emm.q1d <- emm.q1d |>
  mutate(Observation = paste("(x3=", x3, ", ", "x4=", x4, ")", sep=""))
ggplot(emm.q1d) +
  geom_point(aes(x=Mean, y=Observation))+
  geom_errorbar(aes(y=Observation,
                    xmin=Lower, xmax=Upper),
                width = 0.25)+
  # facet_wrap(~x2*x6)+
  theme_bw()+
  xlab("Estimated Mean of Y")
```

Figure 1: Estimated marginal means for the model in Question 1.



1.5 Part e

Untangle and interpret the $x_5 * x_6$ interaction using algebra. Use Table 2 and Figure 2 for context.

```
emm.q1e <- emtrends(mod.q1, var="x5", ~x5|x6) |>
            data.frame() |>
            rename("Slope"  = "x5.trend",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL") |>
            dplyr::select(x6, Slope, SE, df, Lower, Upper)
```

Table 2: Estimated marginal trends for the model in Question 1.

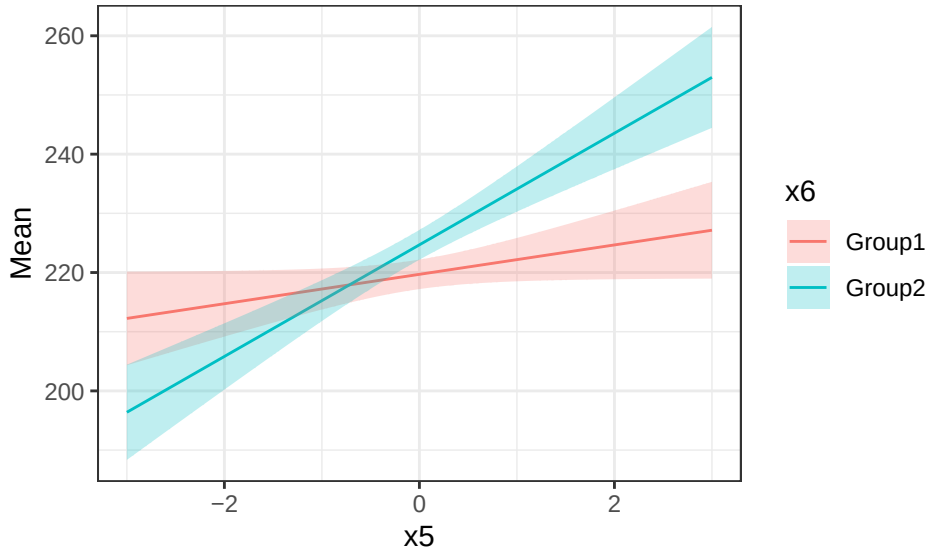
x6	Slope	SE	df	Lower	Upper
Group1	2.4835	1.2942	238	-0.0660	5.0330
Group2	9.4309	1.3367	238	6.7977	12.0642

In Figure 2, the slopes are the same and just the intercepts vary depending on x_2, x_3, and x_4.

```r
emm.q1e <- emmeans(mod.q1, ~x5*x6,
                   at = list(x5 = seq(-3, 3, length.out=500))) |> # Don't average across levels: use |x2*x3*x4
  data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")

ggplot(emm.q1e) +
  geom_line(aes(x=x5, y=Mean, color=x6))+
  geom_ribbon(aes(x=x5, ymin=Lower, ymax=Upper, fill=x6), alpha=0.25)+
  # facet_wrap(~x2*x3*x4)+
  theme_bw()
```

Figure 2: Estimated marginal trends for the model in Question 1.



1.6 Part f

Untangle and interpret the $x_7 * x_8$ interaction using algebra. Use Table 3 and Table 4 for context.

```r
emm.q1f.x7 <- emtrends(mod.q1, var="x7", ~x7|x8,
                   at = list(x8 = mean(dat.q1$x8) + c(-1, 0 ,1)*sd(dat.q1$x8))) |>
            data.frame() |>
            rename("Slope"  = "x7.trend",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL") |>
            dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 3: Estimated marginal trends for the model in Question 1.

x8	Slope	SE	df	Lower	Upper
8.1309	7.9230	0.6751	238	6.5932	9.2529
10.0582	9.7129	0.5015	238	8.7250	10.7008
11.9854	11.5027	0.7396	238	10.0458	12.9597

```r
emm.q1f.x8 <- emtrends(mod.q1, var="x8", ~x8|x7,
                   at = list(x7 = mean(dat.q1$x7) + c(-1, 0 ,1)*sd(dat.q1$x7))) |>
            data.frame() |>
            rename("Slope"  = "x8.trend",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL") |>
            dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 4: Estimated marginal trends for the model in Question 1.

x7	Slope	SE	df	Lower	Upper
8.3824	6.8569	0.6069	238	5.6613	8.0525
10.2208	8.5643	0.4744	238	7.6297	9.4989
12.0592	10.2716	0.7325	238	8.8286	11.7146

A Johnson Neyman analysis using the `interactions` package for `R` (Long 2024), shows that the slope of x_7 is statistically discernible for all observed values of x_8 (Figure 3), and vice versa (Figure 4).

```
library(interactions)
johnson_neyman(mod.q1, pred = x7, modx = x8, control.fdr = T)
```

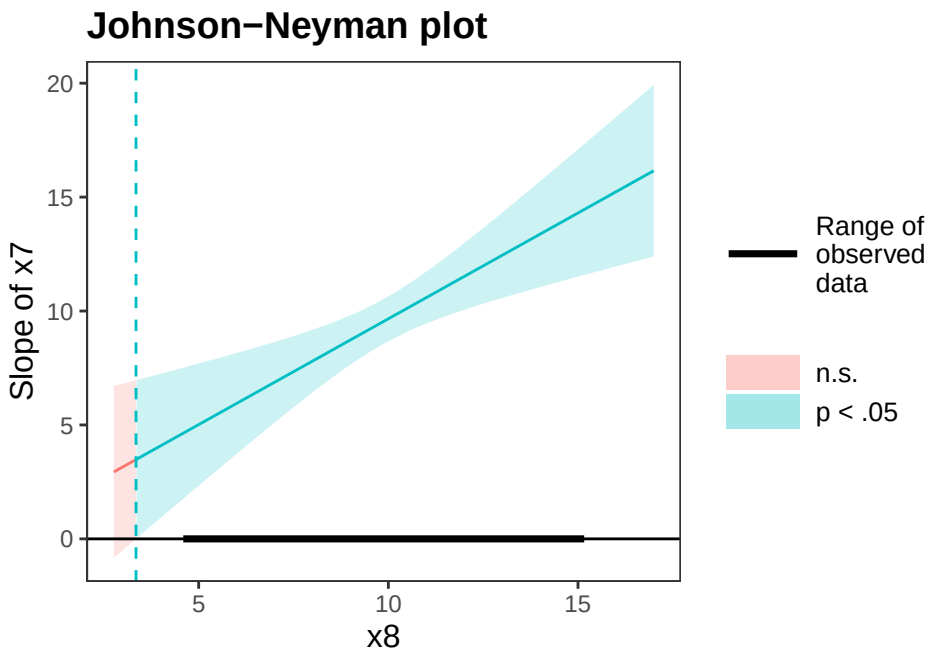
```
JOHNSON-NEYMAN INTERVAL

When x8 is OUTSIDE the interval [-13.08, 3.35], the slope of x7 is p < .05.

Note: The range of observed values of x8 is [4.69, 15.07]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 3: Johnson Neyman interval for the model in Question 1.




```
johnson_neyman(mod.q1, pred = x8, modx = x7, control.fdr = T)
```

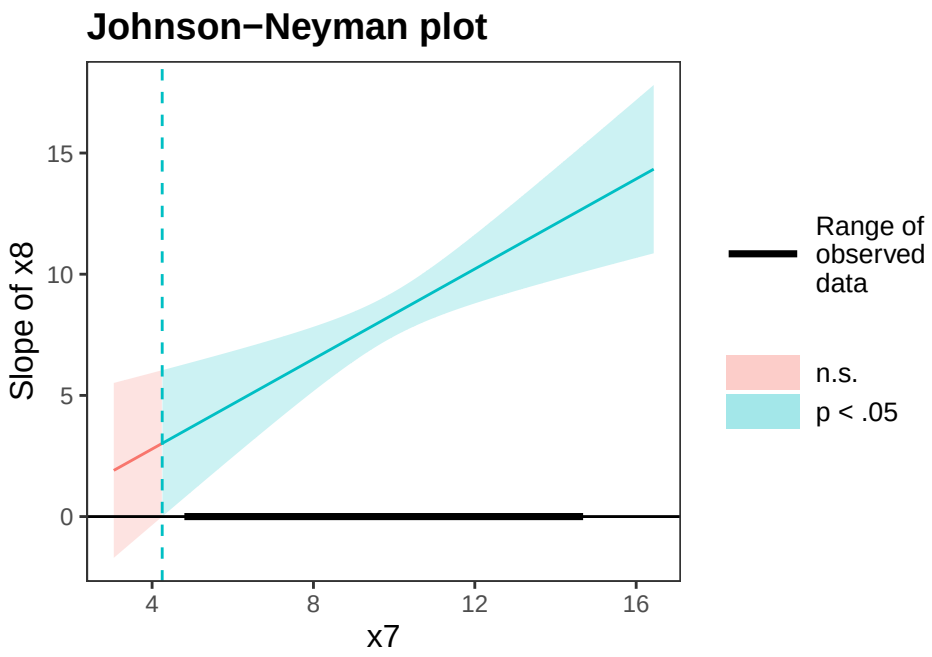
```
JOHNSON-NEYMAN INTERVAL

When x7 is OUTSIDE the interval [-9.96, 4.25], the slope of x8 is p < .05.

Note: The range of observed values of x7 is [4.89, 14.60]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 4: Johnson Neyman interval for the model in Question 1.



1.7 Part g

Standardize the continuous data and refit the regression model. Redo part f. What do you notice about the coefficient estimates and why does this make the untangling process easier?

Solution: First, we fit the regression model using `lm()`.

```
# Standardize Data
dat.z <- dat.q1 |>
  mutate(across(c(x1, x5, x7, x8), ~ as.vector(scale(.))))
# Refit Model
mod.q1z <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.z)
```

```
emm.q1g.z7 <- emtrends(mod.q1z, var="x7", ~x7|x8,
                       at = list(x8 = c(-1, 0 ,1))) |>
          data.frame() |>
          rename("Slope"  = "x7.trend",
                 "Lower" = "lower.CL",
                 "Upper" = "upper.CL") |>
          dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 5: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x8	Slope	SE	df	Lower	Upper
-1	14.57	1.24	238	12.12	17.01
0	17.86	0.92	238	16.04	19.67
1	21.15	1.36	238	18.47	23.83

```
emm.q1g.z8 <- emtrends(mod.q1z, var="x8", ~x8|x7,
                       at = list(x7 = c(-1, 0 ,1)))|>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 6: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x7	Slope	SE	df	Lower	Upper
-1	13.22	1.17	238	10.91	15.52
0	16.51	0.91	238	14.70	18.31
1	19.80	1.41	238	17.01	22.58

2 Question 2

Cui and Mao (2025) investigate how freezing and cryogenic temperatures affect the strength, flexibility, stiffness, and internal breaking behavior of 3D-printed CF/PEEK composites.

Carbon fiber reinforced PEEK (CF/PEEK) is a high-performance composite used in 3D printing for aerospace, deep-sea, polar research. The researchers aimed to assess whether and how extremely cold environments impact this composite to understand how 3D-printed CF/PEEK performs in freezing conditions. This work would be important to understand its limits and develop preventative methods to protect against structural failures.

The researchers created fifteen 3D printed dog-bone-shaped test specimens and placed them in a chamber cooled with liquid nitrogen to 25°C, −25°C, −75°C, −125°C, and −175°C. To test strength of the specimen, a machine pulled both ends of the specimen until it snapped.

In the table below, the researchers record the average tensile strength for three specimens from each temperature level.

Temperature	Mean Tensile Strength
25°C	81.93 MPa
−25°C	76.63 MPa
−75°C	63.23 MPa
−125°C	32.13 MPa
−175°C	12.83 MPa

2.1 Part a

Compute the estimates we derived using calculus in lecture 16:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)^2}.$$

```
dat.q2 = tibble(Temperature = c(25, -25, -75, -125, -175), Strength = c(81.93, 76.63, 63.23, 32.13, 12.83))
sum1 = 0
sum2 = 0
xbar = mean(dat.q2$Temperature)
ybar = mean(dat.q2$Strength)
for(i in 1:length(dat.q2$Temperature)){
  sum1 = sum1 + ((dat.q2$Temperature[i] - xbar)*(dat.q2$Strength[i] - ybar))
  sum2 = sum2 + (dat.q2$Temperature[i] - xbar)^2
}
b1 = sum1/sum2
b0 = ybar - b1*xbar
dat.regression = tibble(method = "Calculus", b_0 = b0, b_1 = b1)
```

Using the calculus approach the regression model we get is $y = 80.755 + 0.3654x_1$.

2.2 Part b

Compute the estimates we derived using linear algebra in lecture 16:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Hint: Run `?matrix()` in `R` for a quick reminder about how to create a matrix.

```
X = matrix(c(rep(1, 5), dat.q2$Temperature), nrow = 5, ncol = 2)
Y = matrix(dat.q2$Strength)

Bhat = solve(t(X)%*%X)%*%t(X)%*%Y
dat.regression = dat.regression |>
  add_row(method = "Linear Algebra", b_0 = Bhat[1], b_1 = Bhat[2])
```

Using the Linear Algebra approach the regression model we get is $y = 80.755 + 0.3654x_1$.

2.3 Part c

Compute the maximum likelihood estimates using `optim()` in `R` as described in lecture 16:

$$\max_{\beta_0,\beta_1,\sigma} \mathcal{L}(\theta = \{\beta_0, \beta_1, \sigma\}).$$

```
y = dat.q2$Strength
x = dat.q2$Temperature
n = length(y)

ll_theta = function(args){
  b0 = args[1]
  b1 = args[2]
  sd = args[3]
  sum = 0
  for(i in 1:n){
    sum = sum + ((y[i]-(b0+b1*x[i]))^2)/(2*sd^2)
  }
  theta = -((-n/2)*log(2*pi)-(n/2)*log(sd^2)-sum)
}

theta = optim(c(78, 0.5, 1), ll_theta)
dat.regression = dat.regression |>
  add_row(method = "Log Likliehood", b_0 = theta$par[1], b_1 = theta$par[2])
```

Using the log likliehood method the computed regression is $y = 80.7449173 + 0.3653252x_1$

2.4 Part d

Fit the model using `lm()` and plot the resulting model. Compare to the results from parts (a) through (c).

```
reg = lm(Strength ~ Temperature, dat.q2)
dat.regression = dat.regression |>
  add_row(method = "`lm()`", b_0 = reg$coefficients[1], b_1 = reg$coefficients[2])
knitr::kable(dat.regression)
```

method	b_0	b_1
Calculus	80.75500	0.3654000
Linear Algebra	80.75500	0.3654000
Log Likliehood	80.74492	0.3653252
`lm()`	80.75500	0.3654000

Comparing the results in the table above, all methods are the same except for the log likliehood estimation. This makes sense because the calculus and linear algebra approaches are exact methods for calculating the regression coeffecients while the log likliehood method is an approximation using computational methods.

3 Question 3

Bracht et al. (2016) evaluate MFAP4 as a biomarker for severity of fibrosis in hepatitic C patients.

```
dat.mfap4 <- read_csv("Bracht2016.csv")
```

3.1 Part a

Fit the model predicting `MFAP4` using `Fibrosis.Stage`. Write out the resulting estimate model and interpret the coefficients.

```
mod.q3 = lm(MFAP4 ~ Fibrosis.Stage, dat.mfap4)
```

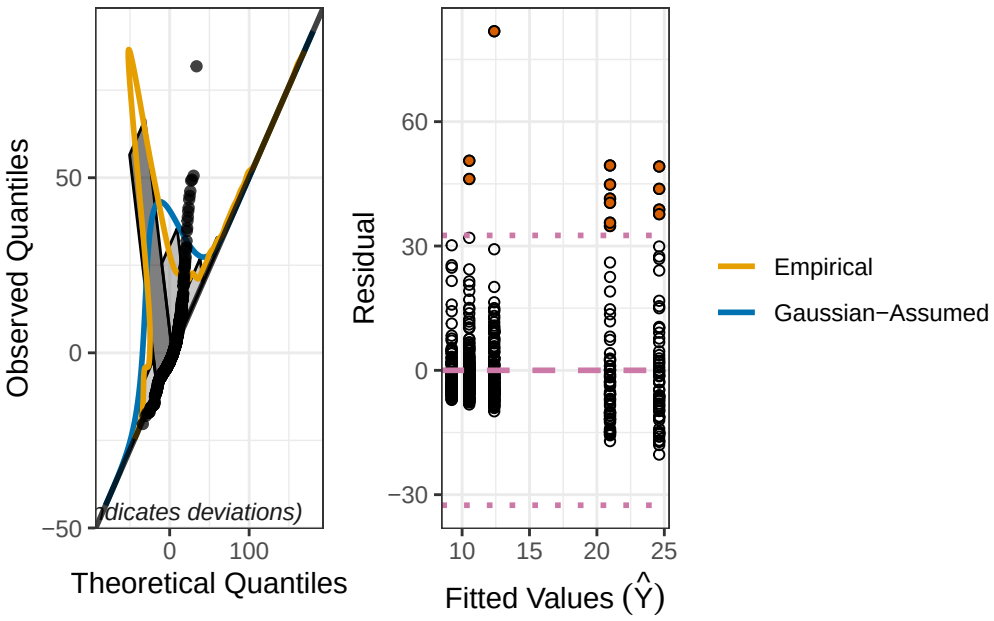
$$\hat{y} = 9.23 + 1.3x_1 + 3.15x_2 + 11.74x_3 + 15.39x_4$$

In the regression model above y represents MFAP4 level, the intercept represents the baseline MFAP4 levels, someone with stage 0 fibrosis. x_1 through x_4 represent the stages of fibrosisis, if the patient has fibrosis stage n then x_n is 1 and the rest are 0. The coefficients represent how much MFAP4 levels are predicted to increase from the intercept (baseline) based on the observed fibrosis stage.

3.2 Part b

Conduct a residual analysis on the model in part a to demonstrate how to identify when you need a log transformation.

```
plotResiduals(mod.q3)
```



The residual plot shows a clear increase of variation as the fibrosis stages increase, which is a sign of a nonlinear relationship, aswell as several outliers. Adittionaly the qqplot on the left shows the deviation increasing exponentially. In this case it makes sense to do a log transformation because the relationship looks like it could be exponential.

3.3 Part c

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

```
1 mod2.q3 = lm(log(MFAP4) ~ Fibrosis.Stage, dat.mfap4)
```

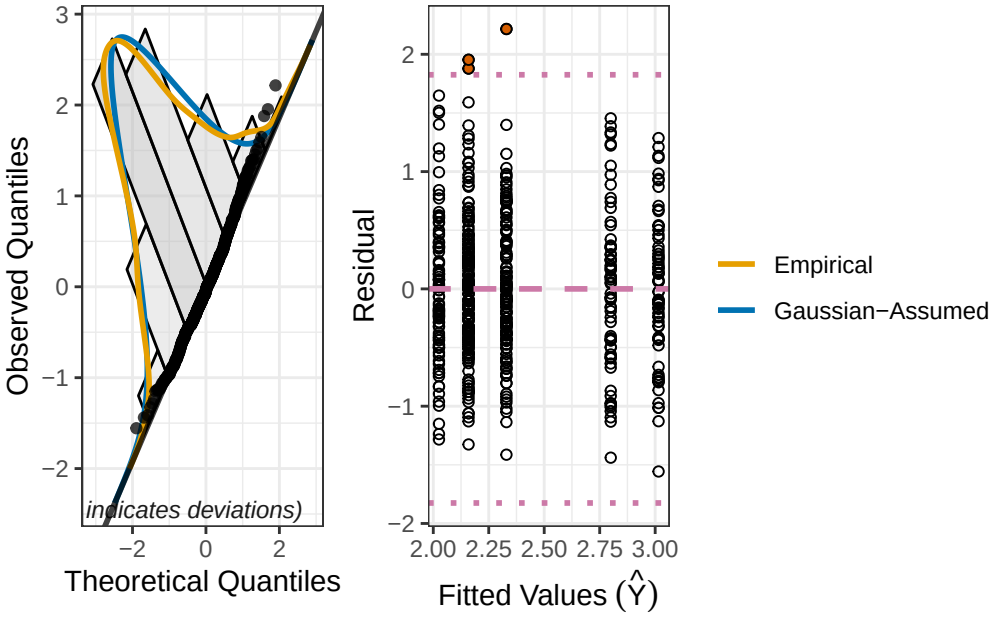
$$\hat{y} = 2.03 + 0.13x_1 + 0.3x_2 + 0.77x_3 + 0.99x_4$$

Now the interpretation of the model is the same as the last one excpet we are predicting $\ln$ MFAP4 not MFAP4.

3.4 Part d

Conduct a residual analysis on the model in part c to demonstrate that the conditions for using regression are met.

```
plotResiduals(mod2.q3)
```



The qqplot on the left for the log transformation model shows that the residuals follow a normal distribution unlike in the model before. The residual plot shows only 3 outliers this time and the variance of the residuals are now constant, evident of a linear relationship. This model much more strongly shows the relationship between MFAP4 and fibrosis stage and more accurately predicts MFAP4 values.

3.5 Part e

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`, `Age`, and `Gender`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

```
mod3.q3 = lm(log(MFAP4) ~ Fibrosis.Stage + Age + Gender, dat.mfap4)
```

$$\hat{y} = 1.52 + 0.13x_1 + 0.21x_2 + 0.6x_3 + 0.83x_4 + 0.01x_5 - 0.14x_6$$

x_1 = Fibrosis Stage 1x_2 = Fibrosis Stage 2x_3 = Fibrosis Stage 3x_4 = Fibrosis Stage 4x_5 = Agex_6 = (Gender == Male)

For this model since we have multiple variables of different types, the intercept is predicted log MFAP4 value when an individual is fibrosis stage 0, age 0, and gender female. The next 4 coefficients are the increase in predicted log MFAP4 value based on fibrosis stage, fibrosis stage n would correspond to x_n being 1 and the other fibrosis variables being 0. The sixth coefficient, β_5, is the increase in predicted log MFAP4 value for an increase of one year of age. The last coefficient, β_6 is the increase (or in this case decrease) in predicted log MFAP4 value if the individual is a male.

4 Question 4

4.1 Part a

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Gaussian}(\mu = 0, \sigma = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

4.2 Part b

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Shifted Exponential}(\lambda = 1, \mu = 0) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Hint: Use the `tolerance::r2exp()` function with `rate=1` and `shift=-1`.

4.3 Part c

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Cauchy}(m = 0, s = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

References

Bracht, Thilo, Christian Mölleken, Maike Ahrens, et al. 2016. "Evaluation of the Biomarker Candidate MFAP4 for Non-Invasive Assessment of Hepatic Fibrosis in Hepatitis C Patients." *Journal of Translational Medicine* 14 (1): 201.

Cui, Mengyang, and Jian Mao. 2025. "Effect of Low Temperature on Mechanical Properties and Fracture Behavior of 3D Printed Short Carbon Fiber Reinforced Polyether Ether Ketone Composites (CF/PEEK)." *Scientific Reports* 15 (1): 36244.

Long, Jacob A. 2024. *Interactions: Comprehensive, User-Friendly Toolkit for Probing Interactions*. https://doi.org/10.32614/CRAN.package.interactions.